

Sinyal ve Sistemler

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System Classification: Memoryless (instantaneous)/with memory(dynamic)

Memoryless system's output at time t only depends on input at time t . Otherwise the system is with memory!

Which of these systems are memoryless?

- ✓

(a)

$x(t) \rightarrow \boxed{S} \rightarrow t + x(t)$
- (b)

$x(t) \rightarrow \boxed{S} \rightarrow x(t + 2) = y(t)$
- (c)

$x(t) \rightarrow \boxed{S} \rightarrow x(t - 1) + x(t)$
- ✓

(d)

$x(t) \rightarrow \boxed{S} \rightarrow e^{x(t)}$
- (e)

$x(t) \rightarrow \boxed{S} \rightarrow x(-t) = y(t)$
- (f)

$x(t) \rightarrow \boxed{S} \rightarrow \int_{-\infty}^{\infty} x(v)dv = y(t)$
- (g)

$x(t) \rightarrow \boxed{S} \rightarrow \int_{-\infty}^{\infty} x(v)dv = y(t)$
- ✓

(h)

$x(t) \rightarrow \boxed{S} \rightarrow x^2(t)$

Hesâdiki sistemlerden hangisi hafızasızdır?

1) $x(t) \rightarrow \boxed{S} \rightarrow t + x(t)$
• Sistem çıkışı, tüm t değerleri için o anki $x(t)$ değerlerine bağlı olduğundan dolayı sistem hafızasızdır.

2) $x(t) \rightarrow \boxed{S} \rightarrow x(t+2) = y(t)$
• Sistem çıkışı, sadece mevcut girişlere bağlı değil, çıktı $x(t+2)$ yani $+2$ birim gelecekteki girişlere bağlıdır. Hafızalıdır.

3) $x(t) \rightarrow \boxed{S} \rightarrow x(t-1) + x(t)$
• Sistem çıkışı, sadece mevcut girişe bağlı değil, bir birim geçmişteki değere de bağlıdır. Hafızalıdır.

4) $x(t) \rightarrow \boxed{S} \rightarrow e^{x(t)}$
• Sistem çıkışı, tüm t değerleri için o anki $x(t)$ değerlerine bağlı olduğundan sistem hafızasızdır.

System Classification: Memoryless (instantaneous)/with memory(dynamic)

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Which of these systems are memoryless?

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- (e) $x(t) \rightarrow \boxed{S} \rightarrow x(-t) = y(t)$
- (f) $x(t) \rightarrow \boxed{S} \rightarrow \int_{-\infty}^{\infty} x(v)dv = y(t)$
- (g) $x(t) \rightarrow \boxed{S} \rightarrow \int_{-\infty}^{\infty} x(v)dv = y(t)$
- ✓ (h) $x(t) \rightarrow \boxed{S} \rightarrow x^2(t)$

5) $x(t) \rightarrow \boxed{S} \rightarrow x(-t) = y(t)$
• Verilen sisteminde $t \geq 0$ olduğundan geçmiş, $t < 0$ olduğunda gelecek $t \geq 0$ olduğunda mevcut değere bağlı olabileceğinden sistem hafızalıdır.

6) $x(t) \rightarrow \boxed{S} \rightarrow \int_{-\infty}^{\infty} x(u)du = y(t)$
• t zamanındaki giriş, girişin eksi sonsuzdan artı sonsuza kadar olan integraline bağlıdır. Hafızalıdır, yani girişin geçmiş ve gelecek değerlerine bağlı.

7) $x(t) \rightarrow \boxed{S} \rightarrow \int_{-\infty}^t x(u)du = y(t)$
• Sistem çıktısı tüm geçmiş değerlerden mevcut t anına kadar toplanan geçen geçmiş değerlere bağlı olduğundan sistem hafızalıdır.

8) $x(t) \rightarrow \boxed{S} \rightarrow x^2(t)$
• Sistem çıktısı sadece mevcut zamanla bağlı olduğundan sistem hafızasızdır.

System Classification: Causal/ Non-Causal Systems

Causal system's output at time t_0 only depends on input values at time t_0 and at times before t_0 , i.e., only depends on $t \leq t_0$.

Which of these systems are causal?

- ✓ (a) $x(t) \rightarrow \boxed{S} \rightarrow t + x(t)$
- (b) $x(t) \rightarrow \boxed{S} \rightarrow x(t + 2) = y(t)$
- ✓ (c) $x(t) \rightarrow \boxed{S} \rightarrow x(t - 1) + x(t)$
- ✓ (d) $x(t) \rightarrow \boxed{S} \rightarrow e^{x(t)} = y(t)$
- (e) $x(t) \rightarrow \boxed{S} \rightarrow x(-t) = y(t)$
- (f) $x(t) \rightarrow \boxed{S} \rightarrow \int_{-\infty}^{\infty} x(v)dv = y(t)$
- ✓ (g) $x(t) \rightarrow \boxed{S} \rightarrow \int_{-\infty}^{\infty} x(v)dv = y(t)$
- ✓ (h) $x(t) \rightarrow \boxed{S} \rightarrow x^2(t)$

Aşağıdakilerden hangisi nedenseldir?

1) $x(t) \rightarrow \boxed{S} \rightarrow t + x(t)$
• Bu sistemin çıkışı t anındaki değere bağlı olduğundan sistem nedenseldir.

2) $x(t) \rightarrow \boxed{S} \rightarrow x(t+2) = y(t)$
• Sistemin t zamanındaki çıkışı $t+2$ zamanındaki girişe bağlı olduğundan dolayı sistem nedensel değildir.

3) $x(t) \rightarrow \boxed{S} \rightarrow x(t-1) + x(t)$
• Sistemin çıkışı hem t anındaki mevcut değere hem de $t-1$ anındaki geçmiş değere bağlı olduğundan dolayı nedenseldir.

4) $x(t) \rightarrow \boxed{S} \rightarrow e^{x(t)} = y(t)$
• Sistemin t anındaki çıkışı, t anındaki giriş değerine bağlı olduğundan dolayı nedenseldir.

System Classification: Causal/ Non-Causal Systems

Causal system's output at time t_0 only depends on input values at time t_0 and at times before t_0 , i.e., only depends on $t \leq t_0$.

Which of these systems are causal?

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- (b) $x(t) \rightarrow \boxed{S} \rightarrow x(t + 2) = y(t)$
- ✓ (c) $x(t) \rightarrow \boxed{S} \rightarrow x(t - 1) + x(t)$
- ✓ (d) $x(t) \rightarrow \boxed{S} \rightarrow e^{x(t)} = y(t)$
- (e) $x(t) \rightarrow \boxed{S} \rightarrow x(-t) = y(t)$
- (f) $x(t) \rightarrow \boxed{S} \rightarrow \int_{-\infty}^{\infty} x(v)dv = y(t)$
- ✓ (g) $x(t) \rightarrow \boxed{S} \rightarrow \int_{-\infty}^{\infty} x(v)dv = y(t)$
- ✓ (h) $x(t) \rightarrow \boxed{S} \rightarrow x^2(t)$

- 5) $x(t) \rightarrow \boxed{S} \rightarrow x(-t) = y(t)$
- Bu sistemin çıkışı gelecek değerlere de bağlı olabileceği için nedensel değildir. Örneğin $t < 0$ için sistem geleceğe bağlı olur.
- 6) $x(t) \rightarrow \boxed{S} \rightarrow \int_{-\infty}^{\infty} x(v)dv = y(t)$
- Çıkış $y(t)$ girişin eksi sonsuzdan artı sonsuza kadar integralidir. Dolayısıyla t anında sonraki girişlerden etkilenmektedir. Çıkış, girişin gelecek değerlerine bağlı olduğu için sistem nedensel değildir.
- 7) $x(t) \rightarrow \boxed{S} \rightarrow \int_{-\infty}^t x(v)dv = y(t)$
- Çıkış $y(t)$ girişin eksi sonsuzdan t anına kadar integralidir. Çıkış t anında sonraki değerlere bağlı olmadığı için nedenseldir.
- 8) $x(t) \rightarrow \boxed{S} \rightarrow x^2(t)$
- Bu sistemin çıkışı $x^2(t)$ yalnızca t anına bağlı olduğundan dolayı sistem nedenseldir.

System Classification: Invertible/ Non-invertible Systems

Which of these systems are invertible?

- (a) $x(t) \rightarrow \boxed{S} \rightarrow t + x(t)$
- ✓ (b) $x(t) \rightarrow \boxed{S} \rightarrow x(t + 2)$
- (c) $x(t) \rightarrow \boxed{S} \rightarrow x(t - 1) + x(t)$
- ✓ (d) $x(t) \rightarrow \boxed{S} \rightarrow e^{x(t)}$
- ✓ (e) $x(t) \rightarrow \boxed{S} \rightarrow x(-t)$
- (f) $x(t) \rightarrow \boxed{S} \rightarrow \int_{-\infty}^{\infty} x(v) dv$
- (g) $x(t) \rightarrow \boxed{S} \rightarrow \int_{-\infty}^{\infty} x(v) dv$
- (h) $x(t) \rightarrow \boxed{S} \rightarrow x^2(t)$

Is the system $y(t)=t+x(t)$ invertible?

we need to consider whether there exists a unique input for every output.

In this case, $y(t)$ depends both on the input signal $x(t)$ and the time variable t . If we know the output $y(t)$, we can't uniquely determine $x(t)$ because there are infinite combinations of $x(t)$ and t that could yield the same $y(t)$. For example, if $x(t) = 1$ and $t = 2$, the output would be $y(t) = 2 + 1 = 3$. However, if $x(t) = 2$ and $t = 1$, we would still get $y(t) = 1 + 2 = 3$.

Since different inputs can lead to the same output, the system described by $y(t) = t + x(t)$ is not invertible.

Is the system $y(t)=x(t+2)$ invertible?

we need to consider whether there exists a unique input for every output.

In this system, $y(t)$ depends solely on the input signal $x(t)$, specifically on the value of x at $t + 2$.

If we know the output $y(t)$, we can uniquely determine $x(t)$ by evaluating $x(t) = y(t - 2)$.

Since each output $y(t)$ corresponds to a unique input $x(t)$, the system is invertible.

Therefore, $y(t) = x(t + 2)$ represents an invertible system.

Is the system $y(t)=x(t-1) + x(t)$ invertible?

we need to consider whether there exists a unique input for every output.
An invertible system is one where different inputs lead to different outputs, and each output corresponds to a unique input.

In the given system, $y(t)$ depends on both $x(t - 1)$ and $x(t)$. If we know $y(t)$, we can't uniquely determine $x(t - 1)$ and $x(t)$ separately because there are multiple combinations of $x(t - 1)$ and $x(t)$ that could yield the same $y(t)$. For example, if $x(t - 1) = 1$ and $x(t) = 2$, the output would be $y(t) = 1 + 2 = 3$. However, if $x(t - 1) = 2$ and $x(t) = 1$, we would still get $y(t) = 1 + 2 = 3$.

Since different inputs can lead to the same output, the system is not invertible.

Is the system $y(t) = e^{x(t)}$ invertible?

we need to consider whether there exists a unique input for every output.

An invertible system is one where different inputs lead to different outputs, and each output corresponds to a unique input.

The system described by $y(t) = e^{x(t)}$ seems to represent an exponential function where the output $y(t)$ depends on the exponential of the input signal $x(t)$.

In this case, if we know the output $y(t)$, we can determine $x(t)$ by taking the natural logarithm of $y(t)$, as $x(t) = \ln(y(t))$.

This means that each output $y(t)$ corresponds to a unique input $x(t)$, given that the natural logarithm function is a one-to-one mapping.

Therefore, the system described by $y(t) = e^{x(t)}$ is invertible.

Is the system $y(t) = x(-t)$ invertible?

we need to consider whether there exists a unique input for every output.

An invertible system is one where different inputs lead to different outputs, and each output corresponds to a unique input.

To determine if the system described by $y(t) = x(-t)$ is invertible, we need to examine whether each output $y(t)$ corresponds to a unique input $x(t)$.

In this system, the output $y(t)$ depends on the input signal $x(t)$ evaluated at the negative time instant $-t$. If we know the output $y(t)$, we can uniquely determine $x(t)$ by evaluating $x(t) = y(-t)$.

Since each output $y(t)$ corresponds to a unique input $x(t)$, the system is invertible.

Therefore, $y(t) = x(-t)$ represents an invertible system.

Is the system $y(t) = \int_{-\infty}^{\infty} x(v) dv$ invertible?

In general, the integral $\int_{-\infty}^{+\infty} x(v) dv$ over the entire real line from negative infinity to positive infinity may not be invertible.

The reason for this is that the integral operation aggregates information from the entire domain of v into a single value. If two different functions $x_1(v)$ and $x_2(v)$ have the same integral value over the entire real line, then the integral operation does not provide enough information to uniquely determine the original functions.

For example, consider two functions $x_1(v)$ and $x_2(v)$ such that:

1. $x_1(v) = e^{-v^2}$ (Gaussian function).
2. $x_2(v) = -e^{-v^2}$ (Negated Gaussian function).

Both of these functions have the same integral value over the entire real line:

$$\begin{aligned}\int_{-\infty}^{+\infty} e^{-v^2} dv &= \sqrt{\pi} \\ \int_{-\infty}^{+\infty} -e^{-v^2} dv &= -\sqrt{\pi}\end{aligned}$$

Therefore, the integral operation alone cannot uniquely determine the original function.

However, if additional information or constraints are provided, such as specific properties of $x(v)$ or the context in which it is used, it may be possible to reconstruct the original function from its integral. In such cases, the operation might be considered invertible within the given context.

Is the system $y(t) = \int_{-\infty}^t x(v) dv$ invertible?

The integral from negative infinity to t of $x(v) dv$, denoted as $\int_{-\infty}^t x(v) dv$, is not generally invertible.

Here's why:

1. **Integral Operation:** This operation aggregates the information of the function $x(v)$ over the interval from negative infinity to t into a single value.
2. **Lack of Information:** Just like the integral over the entire real line, if two different functions $x_1(v)$ and $x_2(v)$ have the same integral value from negative infinity to t , then the operation does not provide enough information to uniquely determine the original functions.

For example, consider two functions $x_1(v)$ and $x_2(v)$ such that:

$$x_1(v) = e^{-v^2} \text{ (Gaussian function)}$$

$$x_2(v) = -e^{-v^2} \text{ (Negated Gaussian function)}$$

Both of these functions have the same integral value from negative infinity to t for any t :

$$\int_{-\infty}^t e^{-v^2} dv = \frac{\sqrt{\pi}}{2}$$

$$\int_{-\infty}^t -e^{-v^2} dv = -\frac{\sqrt{\pi}}{2}$$

Thus, the integral operation alone cannot uniquely determine the original function $x(v)$.

However, similar to the case of the integral over the entire real line, if additional information or constraints are provided, such as specific properties of $x(v)$ or the context in which it is used, it may be possible to reconstruct the original function from its integral over the interval from negative infinity to t . In such cases, the operation might be considered invertible within the given context.

Is the system $y(t)=x^2(t)$ invertible?

To determine if the system described by $y(t) = x^2(t)$ is invertible, we need to check if each output $y(t)$ corresponds to a unique input $x(t)$.

In this system, the output $y(t)$ depends on the square of the input signal $x(t)$. Given an output $y(t)$, there could be multiple inputs $x(t)$ that produce the same output because squaring a number may result in the same output for different inputs (e.g., $x(t) = 2$ and $x(t) = -2$ both yield $y(t) = 4$).

Therefore, since different inputs can lead to the same output, the system described by $y(t) = x^2(t)$ is not invertible.

System Classification: Stable/Unstable Systems

Which of these systems are externally stable?

- (a) $x(t) \rightarrow \boxed{S} \rightarrow t + x(t)$
- ✓ (b) $x(t) \rightarrow \boxed{S} \rightarrow x(t + 2)$
- (c) $x(t) \rightarrow \boxed{S} \rightarrow x(t - 1) + x(t)$
- ✓ (d) $x(t) \rightarrow \boxed{S} \rightarrow e^{x(t)}$
- ✓ (e) $x(t) \rightarrow \boxed{S} \rightarrow x(-t)$
- (f) $x(t) \rightarrow \boxed{S} \rightarrow \int_{-\infty}^{\infty} x(v) dv$
- (g) $x(t) \rightarrow \boxed{S} \rightarrow \int_{-\infty}^{\infty} x(v) dv$
- ✓ (h) $x(t) \rightarrow \boxed{S} \rightarrow x^2(t)$

Is the system $y(t)=t+x(t)$ stable?

The stability of a system $y(t) = t + x(t)$ can be assessed using the bounded-input bounded-output (BIBO) stability criterion. According to this criterion, if the system is stable, then for any bounded input signal, the output remains bounded.

In this system, the output $y(t)$ depends on the input signal $x(t)$ and the time variable t . The time variable t grows unbounded as time progresses, so the output $y(t)$ will also grow unbounded over time, regardless of the behavior of the input signal $x(t)$.

Since the output can become unbounded even for bounded inputs, the system $y(t) = t + x(t)$ is not stable according to the BIBO stability criterion.

Is the system $y(t)=x(t+2)$ stable?

The system $y(t) = x(t + 2)$ is a time-shifted version of the input signal $x(t)$. In this case, the time shift by $+2$ units doesn't inherently introduce instability because it's a simple time shift.

To assess stability, we often consider the bounded-input bounded-output (BIBO) stability criterion. According to this criterion, if the system is stable, then for any bounded input signal, the output remains bounded.

For the system $y(t) = x(t + 2)$, if the input signal $x(t)$ is bounded for all t , then $y(t)$ will also remain bounded for all t . This is because the time shift doesn't affect the boundedness of the input signal; it only changes its timing.

Therefore, assuming that the input signal $x(t)$ is bounded for all t , the system $y(t) = x(t + 2)$ can be considered stable according to the BIBO stability criterion.

Is the system $y(t)=x(t-1) + x(t)$ stable?

In this system, the output $y(t)$ is the sum of two terms: $x(t - 1)$ and $x(t)$. A bounded input signal $x(t)$ doesn't guarantee that $x(t - 1)$ will also be bounded, especially if t extends to negative infinity.

For instance, consider a bounded input signal $x(t)$ that is bounded in the interval $[0, \infty)$. However, $x(t - 1)$ is not bounded, especially when t approaches negative infinity.

As a result, the system $y(t) = x(t - 1) + x(t)$ may not always yield a bounded output for bounded inputs. Therefore, this system may not be stable according to the BIBO stability criterion.

Is the system $y(t) = e^{x(t)}$ stable?

In the case of $y(t) = e^{x(t)}$, if the input signal $x(t)$ is bounded, then $e^{x(t)}$ will generally produce bounded output. This is because the exponential function $e^{x(t)}$ grows rapidly as $x(t)$ becomes large in magnitude, but it does not grow without bound for bounded $x(t)$.

Therefore, under the assumption that the input signal $x(t)$ is bounded for all t , the system $y(t) = e^{x(t)}$ can be considered stable according to the BIBO stability criterion.

Is the system $y(t) = x(-t)$ stable?

In the given system, the output $y(t)$ is obtained by evaluating the input signal $x(t)$ at the negative time $-t$. This essentially reflects the input signal about the origin (time reversal).

If the input signal $x(t)$ is bounded for all t , then $x(-t)$ will also be bounded since the operation of reflecting about the origin does not alter the boundedness of the signal.

Therefore, under the assumption that the input signal $x(t)$ is bounded for all t , the system $y(t) = x(-t)$ can be considered stable according to the BIBO stability criterion.

Is the system $y(t) = \int_{-\infty}^{\infty} x(v) dv$ stable?

In the context of systems theory, the concept of stability typically refers to the behavior of a system's output given certain inputs. However, for the integral operation $\int_{-\infty}^{+\infty} x(v) dv$, it doesn't directly relate to stability in the same way as a system with inputs and outputs.

The integral operation essentially accumulates the area under the curve of the function $x(v)$ over the entire real line from negative infinity to positive infinity. Whether this operation is considered "stable" depends on the properties of the function $x(v)$ and the context in which it is used.

If the function $x(v)$ is bounded for all v , then the integral operation will also produce a bounded output, as it accumulates only bounded values.

However, if $x(v)$ is unbounded (i.e., grows without bound) over certain intervals, then the integral operation may result in an unbounded output. In this case, the operation could be considered unstable in the sense that it may produce unbounded output for certain bounded inputs.

Therefore, whether the integral operation $\int_{-\infty}^{+\infty} x(v) dv$ is considered "stable" depends on the properties of the function $x(v)$ and the context in which it is used.

Is the system $y(t) = \int_{-\infty}^t x(v) dv$ stable?

The concept of stability for an integral operation like $\int_{-\infty}^t x(v) dv$ is different from that of a system with inputs and outputs. In the context of systems theory, stability typically refers to the behavior of the output given a certain input.

However, for the integral operation you've mentioned, it doesn't make sense to talk about stability in the same way. Instead, we can discuss whether the integral operation introduces any instability or causes the output to become unbounded under certain conditions.

The integral operation essentially accumulates the area under the curve of the function $x(v)$ from negative infinity up to the time t . If the function $x(v)$ is bounded for all v , then the integral operation will also produce a bounded output for any t , as it accumulates only bounded values.

However, if $x(v)$ is unbounded (i.e., grows without bound) over certain intervals, then the integral operation may result in an unbounded output for specific values of t . In this case, the operation could be considered unstable in the sense that it may produce unbounded output for certain bounded inputs.

Therefore, whether the integral operation $\int_{-\infty}^t x(v) dv$ is considered "stable" depends on the properties of the function $x(v)$ and the context in which it is used.

Is the system $y(t)=x^2(t)$ stable?

The stability of the system $y(t) = x^2(t)$ can be evaluated based on how the output $y(t)$ behaves with respect to bounded inputs $x(t)$.

To assess stability, we typically employ the Bounded-Input Bounded-Output (BIBO) stability criterion. According to this criterion, if the system is stable, then for any bounded input signal, the output remains bounded.

In the case of $y(t) = x^2(t)$, if the input signal $x(t)$ is bounded, then squaring $x(t)$ will generally produce a bounded output. This is because squaring a bounded number (or signal) does not cause it to become unbounded.

Therefore, assuming that the input signal $x(t)$ is bounded for all t , the system $y(t) = x^2(t)$ can be considered stable according to the BIBO stability criterion.